

The entire trading pipeline may be written as:

`Trade = Execution(Signal)`

Algorithmic trading makes it possible to react quickly to changes in the market and lessens human emotional bias.

Typically, an algorithmic trading system is made up of several parts that collaborate to create and implement trading strategies.

Market data acquisition

Acquiring trustworthy market data is the first stage in algorithmic trading. This comprises:

- Historical pricing information
- Current market information
- Details about the order book
- Financial and economic indicators

OHLCV values are typically found in market data, where:

- O: Open
- H: High
- L: Low
- C: Close
- V: Volume

Technical analysis and quantitative modelling are based on these characteristics.

Data preprocessing

Preprocessing is required prior to analysis of financial data as raw financial data often contains missing values, irregular timestamps, or outliers.

Typical preprocessing procedures consist of:

- Managing values that are missing
- Normalisation of data and feature engineering
- Alignment of time series

For example, a moving average indicator is computed as:

$$SM A_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

...where P_t is the price at time t and n is the window length.

Signal generation

Mathematical models or predetermined rules are used to produce trading signals. These regulations could be based on:

- Technical indicators
- Models of statistics
- Algorithms for machine learning
- Models with several factors

A straightforward moving average crossover approach, for instance, produces a buy signal when:

$$M A_{short} > M A_{long}$$

....and a sell signal when:

$$M A_{short} < M A_{long}$$

This type of strategy attempts to capture market trends.

Risk management

An essential part of algorithmic trading is risk control. If risks are not well managed, even winning tactics might result in large losses. Important risk management strategies consist of:

- Stop-loss orders
- Sizing of positions
- Diversification of portfolios
- Control of maximum drawdown

The following is an example of a basic stop-loss rule:

$$Loss \leq Threshold$$

If the loss exceeds the threshold, the position is automatically closed.

Order execution

The system uses an exchange API or brokerage interface to send orders to the market once a trading signal has been created. Algorithms for execution seek to reduce:

- Market impact
- Transaction costs
- Slippage

Examples of execution algorithms include:

- Time Weighted Average Price (TWAP)
- Volume Weighted Average Price (VWAP)
- Smart Order Routing (SOR)

Backtesting

Backtesting uses past market data to assess a trading strategy's performance. Before implementing techniques in actual markets, it enables traders to assess possible risks and rewards.

Key performance indicators consist of:

- Total return
- Sharpe ratio
- Maximum drawdown
- Win-loss ratio

The definition of the Sharpe ratio is:

$$Sharpe = \frac{R_p - R_f}{\sigma_p}$$

...where:

- R_p = Portfolio return
- R_f = Risk-free rate
- σ_p = Standard deviation of portfolio returns